MIL-M-20482D(MU)
26 FEBRUARY 1965
SUPERSEDING
MIL-M-20482C(MU)
29 APRIL 1964

MILITARY SPECIFICATION

MOTOR, ROCKET, M95

PARTS FOR

1. SCOPE

1.1 This specification contains requirements not required by the drawings and provides for quality assurance provisions for the fabrication of parts, assembling and packing for one type of rocket motor designated as Motor, Rocket, M95 (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-R-11470 — Radiographic Inspection; Qualification of Equipment, Operators and Procedures.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

FEDERAL

Fed Std 151 — Metals, Test Methods.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-435 — Inspection, Radiographic.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8798125 — Box, Packing, Ammunition, for Rocket Motor M95.

8800014 — Rocket Motor, M95, Metal Parts Assembly.

FSC 1440

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PUBLICATIONS

ORDNANCE CORPS

IEL–8800014 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (dwg.) 8800014 and with all requirements specified in applicable specifications.

3.3 Welds. All welds shall be complete and shall show no evidence of cracks or separations in the weld metal or base metal. They shall be of uniform size, width, smoothness and reinforcement and shall be free from craters, slag pockets, porosity, cracks, undercuts, and other surface imperfections.

3.4 Strength of adapter weld. The adapter weld in the motor body assembly shall withstand application of a tensile force of 6000 pounds, minimum, with no evidence of weld failure.

3.5 Workmanship. The item of shipment, its components, and materials shall be free of dirt, chips, grease, corrosions, and other foreign material. There shall be no cracks, scratches, burrs, dents, gouges, holes, porous areas, distortions, or other undesirable characteristics which would, in any way, detract from the useability of the product for its intended purpose. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The painting and coating on all components shall be restored when such paintings and coatings have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specification referenced in any of the contractual documents.

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(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide. the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* The contractor shall submit 20 assemblies and 10 complete sets of parts (i.e. 10 each of every individual component and every subassembly) with required packing, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production quantity shall be accompanied by certificates of analysis for all materials. An initial production quantity shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, or specification, such as to significantly affect product uniformity as determined by the Government. After successful submission to examination and test, the 20 units submitted as complete assemblies shall be forwarded to the loading contractor for use in the initial production sample.

4.2.2 *Inspections to be performed.* Each component, subassembly, and assembly shall be inspected for every dimension, and requirement of the applicable drawing and 4.3.3.1.

4.2.3 *Rejection.* If any assembly, subassembly or component fails to comply with all applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until an acceptable sample is submitted or until otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be ap-

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plied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Manifold after hydrostatic test, and prior to painting (see dwg. 8800015 covering a detail of dwg. 8800014).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:			
1.	Hydrostatic test stamp missing	Visual	01001
Major:	AQL 1.00 percent		
101.	Location of nozzle thread	Gage	01002
102.	Pitch diameter of nozzle thread maximum (max.)	Gage	01003
103.	Wall thickness below nozzle thread including conical flat, and cylindrical sections, minimum (min.)	Gage	01004
104.	Evidence of poor workmanship on any interior surface (see 3.5)	Visual	01005
105.	Concentricity of pitch diameter with diameter before motor body thread	Gage	01006
106.	Perpendicularity of front trap shoulder to diameter before motor body thread	Gage	01007
Minor:	AQL 4.00 percent		
201.	Pitch diameter of nozzle thread, min.	Gage	01008
202.	Minor diameter of nozzle thread, max.	Gage	01009
203.	Length of nozzle thread, min.	Gage	01010
204.	Pitch diameter of motor body thread, min.	Gage	01011
205.	Minor diameter of motor body thread, max.	Gage	01012
206.	Length of motor body thread, min.	Gage	01013
207.	Outside diameter, max.	Gage	01014
208.	Depth to shoulder behind thread	Gage	01015

4.3.2.2 Manifold (see dwg. 800015 covering a detail of dwg. 8800014).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Rust preventive compound missing or incomplete on threads	Visual	02001
202.	Protective coating damaged exposing base metal	Visual	02002

4.3.2.3 Nozzle (prior to painting) (see dwg. 8800026 covering a detail of dwg. 8800014).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Diameter of throat	Gage	03001
102.	Inside diameter at threaded end	Gage	03002
103.	Inside diameter at tail end	Gage	03003
104.	Pitch diameter of thread, min.	Gage	03004
Minor:	AQL 2.50 percent		
201.	Pitch diameter of thread, max.	Gage	03005
202.	Major diameter of thread, min.	Gage	03006
203.	Length of thread, min.	Gage	03007
204.	Distance to seating surface	Gage	03008
205.	Total length	Gage	03009
206.	Evidence of poor workmanship (see 3.5)	Visual	03010

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4.3.2.4 Nozzle (see dwg. 8800026 covering a detail of dwg. 8800014).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Protective coating damaged exposing base metal (other than slight scratches)	Visual	04001
202.	Rust preventive compound missing or incomplete on thread	Visual	04002
203.	Evidence of poor workmanship (see 3.5)	Visual	04003

4.3.2.5 Front trap assembly (see dwg. 8800020 covering a detail of dwg. 8800019).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Parallelism of strip shoulder	Gage	05001
102.	Perpeendicularity of resonance rod to strip shoulder	Gage	05002
103.	Diameter formed by strip ends	Gage	05003
104.	Weld inadequate (see 3.3)	Visual	05004
Minor:	AQL 2.50 percent		
201.	Pitch diameter of thread, min.	Gage	05005
202.	Distance trap ring to end of rod, max.	Gage	05006
203.	Protective coating missing or inadequate	Visual	05007
204.	Compound missing from threads	Visual	05008
205.	Evidence of poor workmanship (see 3.5)	Visual	05009

4.3.2.6 Rear trap (see dwg 8800021 or 8833708 (as applicable) covering a detail of dwg. 8800019).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pitch diameter of center therad, max.	Gage	06001
102.	Pitch diameter of leg thread, max.	Gage	06002
103.	Flatness of igniter bearing surface	Gage	06003
104.	Location of end hole	Gage	06004
Minor:	AQL 1.50 percent		
201.	Diameter formed by leg ends	Gage	06005
202.	Protective coating missing or inadequate	Visual	06006
203.	Compound missing from thread	Visual	06007
204.	Evidence of poor workmanship (see 3.5)	Visual	06008

4.3.2.7 Trap assembly (see dwg. 8800019 covering a detail of dwg. 8800014).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Component missing or loose	Visual-Manual	07001
Minor:	AQL 1.00 percent		
201.	Protective coating damaged	Visual	07002
202.	Evidence of poor workmanship (see 3.5)	Visual	07003

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4.3.2.8 Motor body assembly after hydrostatic test, scale removal and prior to painting (see dwg. 8800016 covering a detail of dwg. 8800014).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Pitch diameter of adapter thread	Gage	08001
102.	Concentricity of adapter thread to outside diameter	Gage	08002
103.	Length of manifold thread, min.	Gage	08003
104.	Operation missing	Visual	08004
105.	Fails to chamber fully	Gage	08005
106.	Concentricity of inside diameter with manifold thread pitch diameter	Gage	08006
Minor:	AQL 2 50 percent		
201.	Distance end to manifold shoulder, min.	Gage	08007
202.	Major diameter of adapter thread	Gage	08008
203.	Length of adapter thread, min.	Gage	08009
204.	Major diameter of manifold thread	Gage	08010
205.	Inside diameter of body	Gage	08011
206.	Evidence of poor workmanship (see 3.5)	Visual	08012

4.3.2.9 Motor body assembly (see dwg. 8800016 covering a detail of dwg. 880014).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Hydrostatic test stamp missing	Visual	09001
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Surface coating inadequate	Visual	09002
202.	Compound missing from thread	Visual	09003

4.3.2.10 Metal parts assembly (see dwg. 8800014).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Component missing	Visual-Manual	10001
102.	Adapter boss thread damaged	Visual	10002
Minor:	AQL 2.50 percent		
201.	Surface coating damaged	Visual	10003
202.	Rust preventive compound incomplete or inadequate on adapter boss thread	Visual	10004
203.	Evidence of poor workmanship (see 3.5)	Visual	10005
204.	Marking missing, misleading or unidentifiable	Visual	10006

4.3.2.11 Box, packing (see dwg. 8798125).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Strapping missing or broken	Visual	11001
102.	Hardware missing or broken	Visual	11002
103.	Board broken	Visual	11003
104.	Strapping loose	Manual	11004
105.	Hardware loose	Manual	11005

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Contents loose	Manual	11006
202.	Strapping improperly assembled	Visual	11007
203.	Hardware improperly engaged	Visual	11008
204.	Car seal missing, unsealed or improperly positioned	Visual	11009
205.	Marking incorrect, incomplete or illegible	Visual	11010

4.3.3 *Testing.*

4.3.3.1 *Initial production (see 4.2.1).* The samples submitted as the initial production quantity shall be tested as specified below after having been examined for all drawing dimensions and requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Impact test of adapter weld (see dwg. 8800016 and 4.4.1) when applicable — Code No. 08013.

4.3.3.1.2 Pull test of adapter weld (see 3.4 and 4.4.2) — Code No. 08014.

4.3.3.1.3 Hydrostatic test of manifold (see dwg. 8800015 and 4.4.3) — Code No. 01016.

4.3.3.1.4 Hydrostatic test of motor body assembly (see dwg. 8800016 and 4.4.4) — Code No. 08015.

4.3.3.1.5 Yield strength and elongation (see dwg. 8800016 and 4.4.5) — Code No. 08016.

4.3.3.1.6 Radiographic examination of manifold castings (see dwg. 8800015 and 4.4.6) — Code No. 01017.

4.3.3.2 *Production testing.*

4.3.3.2.1 *Impact test of adapter weld when applicable (see dwg. 8800016)— Major defect — Code No. 08013.* The sampling plan and procedures for this test shall be in accordance with MIL-STD-105 using an AQL of 0.25 percent. The test shall be performed in accordance with paragraph 4.4.1 using test equipment specified in 4.3.4. Any assembly which fails to comply with the applicable drawing requirement shall be classed defective.

4.3.3.2.2 *Full test of adapter weld (see 3.4) — Major defect — Code No. 08014.* This test shall be performed 100 percent using the required test equipment of 4.3.4 as specified in 4.4.2. Any assembly containing a weld which exhibits any evidence of failure shall be classed defective and removed from the lot.

4.3.3.2.3 *Hydrostatic test of manifold (see dwg. 8800015) — Critical defect — Code No. 01016.* This test shall be performed 100 percent. The test shall be performed as specified in 4.4.3 using test equipment specified in 4.3.4. Any assembly which fails to comply with the specified drawing shall be classed defective and removed from the lot.

4.3.3.2.4 *Hydrostatic test of motor body assembly (see dwg. 8800016) — Critical defect — Code No. 08015.* This test shall be performed 100 percent. The test shall be performed as specified in 4.4.4 using test equipment specified in 4.3.4. Any assembly which fails to comply with the specified drawing requirement shall be classed defective and removed from the lot.

4.3.3.2.5 *Yield strength and elongation (see dwg. 8800016) — Major defect — Code No. 08016.* A sample of ten (10) motor body assemblies shall be selected at random from each lot for this test. The test shall be conducted as specified in 4.4.5. If any motor body assembly fails to meet the applicable drawing requirements, the lot shall be rejected.

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4.3.3.2.6 *Radiographic examination of manifold castings (see dwg. 8800015) — Major defect — Code No. 01017.* The sampling plan and procedures for this test shall be in accordance with MIL–STD–105 using an AQL of 0.25 percent. The test shall be conducted as specified in 4.4.6 using test equipment in accordance with 4.3.4.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8800014 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4. Test methods and procedures.

4.4.1 *Impact test of adapter weld.* When applicable the motor body assembly shall be assembled to the appropriate test equipment specified in 4.3.4. The force specified on the applicable drawing shall be applied axially to the adapter weld in less than $\frac{1}{20}$ second. After the test the weld shall be examined for evidence of failure. Any assembly exhibiting such a defect shall be classed defective and removed from the lot. This test shall be performed prior to the pull test of the adapter weld.

4.4.2 *Pull test of adapter weld.* Prior to hydrostatic testing, the motor body assembly shall be assembled to the applicable test equipment and the required force applied perpendicular to the adapter weld. The force shall be applied for 10 seconds, minimum. After the test, the weld shall be examined by the magnetic particle method for evidence of failure. Any assembly exhibiting such a defect shall be classed defective.

4.4.3 *Hydrostatic test of manifold.* The unpainted manifold shall be assembled in the applicable test fixture specified in 4.3.4, in such a manner as to insure that the manifold is unsupported except at the motor body threads. The manifold shall then be subjected to the hydrostatic pressure specified on the applicable drawing for a minimum of ten seconds. Water or oil having a viscosity which does not exceed 200 Saybolt Universal seconds at 100 degrees Fahrenheit (F.) shall be used in this test. (If water is used, it shall contain a bright coloring agent to facilitate the visual detection of leaks. Extreme care shall be taken to insure that the manifold is thoroughly dried after testing to prevent rusting.) Observation shall be made for evidence of leakage cracks and permanent distortion in excess of drawing dimensions, including threads. Permanent distortion shall be determined by gaging the pitch diameter of the motor body thread and the symmetry of the nozzles with the diameter before the motor body thread before and after the hy-

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drostatic testing and observation made for compliance with the drawing requirements.

4.4.4 *Hydrostatic test of motor body assembly.* The unpainted motor body shall be assembled to the applicable test fixture specified in 4.3.4 in such a manner as to insure that the motor body is unsupported except at the manifold threads. In this position the motor body shall be subjected to the hydrostatic pressure specified on the applicable drawing for a minimum of ten seconds. Water or oil having a viscosity which does not exceed 200 Saybolt Universal seconds at 100 degrees F. shall be used in this test. If water is used, it shall contain a bright coloring agent to facilitate the visual detection of leaks. Extreme care shall be taken to insure that the motor body is thoroughly dry, after testing, to prevent rusting. Observation shall be made for evidence of leakage, cracks, and permanent distortion in excess of drawing dimensions, including threads. Permanent distortion shall be determined by gaging the pitch diameter of the manifold thread and the length from the end to bottom of radius before and after the hydrostatic test and observation made for compliance with the drawing requirements.

4.4.5 *Yield strength and elongation.* This test shall be conducted in accordance with Federal Methods Standard 151, Method 211.1. Two longitudinal test specimens taken 90 degrees apart shall be prepared. One shall be taken near the mid-section of the motor body assembly and the other taken near the manifold thread. Observation shall be made for compliance with the applicable requirements on the drawing.

4.4.6 *Radiographic examination of manifold castings.* The sample castings shall be subjected to a radiographic examination. The manifold shall be placed on a flat surface with the nozzle cavities down. The first examination shall be taken with the radiographic beam perpendicular to a plane through the nozzles. The second examination shall be taken perpendicular to the flat surface along the longitudinal axis of the manifold. Casting defects shall not exceed those specified in ASTM E 71-52 Radiographic Standards Class 1. Radiographic inspection and qualification of equipment, operators and procedures shall be in accordance with Specification MIL-R-11470. The penetrameter required in MIL-R-11470 shall be fabricated in accordance with MIL-STD-453 to assure a 2 percent minimum sensitivity. The film density shall be no less than 1.25.

5. PREPARATION FOR DELIVERY

5.1. Packing.

5.1.1 *Level C.* The rocket motor assembly shall be packed in accordance with dwg. 8798125.

5.2 Marking. The box shall be marked in accordance with dwg. 8798125.

5.3 Data cards. Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspec-

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tion equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL-D-45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and nonmandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) or their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment.

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This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partia submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 **Submission of initial production samples.** Instructions, as to the location for the evaluation of the Initial Production Sample shall be obtained from the contracting officer Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. Al inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

6.5 **Former designation.** Prior to typ standardization the item of this specification was designated Rocket Motor T4E4.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1340-A-184

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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DEPARTMENT OF THE ARMY

COMMANDING OFFICER
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